

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – CODING CHALLENGE

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 3 – Coding Challenge
Weightage:	30% of CIA		
CO5 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	N.Adi Narayana	Reg. No.:	192525346
Section / Batch:	AIML	Date:	16/7/2026

Code of Conduct

I NAGENDRA REDDY / 192472399 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: NAGENDRA REDDY

Instructions

- Develop a complete and modular Python program that meets all the functional requirements specified in the coding challenge using appropriate algorithms, data structures, and programming practices.
- Test the program using multiple valid and invalid test cases, and clearly display the input, processing, and output to verify the correctness of the implementation.
- Follow standard coding practices by using meaningful variable names, proper indentation, comments, exception handling (where applicable), and upload the source code, sample outputs, and README file to the designated GitHub repository.
- Duration: 40 minutes.

Section / Topic	Focus	Q Nos
Regular Expression Pattern Matching	Regular Expressions, Basic Regular Expression Patterns	1
Finite State Automata Implementation	Finite State Automata, Models and Algorithms	2
Advanced Regular Expression Search	Regular Expressions, Basic Regular Expression Patterns	3

Section 1: Intelligent Email and Password Validator using Regular Expressions

1. Problem Statement:

A software company is developing a user registration system. Before creating an account, the system must validate user credentials based on predefined security policies.

Develop a Python application that validates:

Email Address

Password

Mobile Number

using Regular Expressions.

Validation Rules

Email

Must start with an alphabet.

May contain letters, digits, '.', '_' before '@'.

Domain should contain only alphabets.

Valid extensions: .com, .org, .edu, .net, .in

Password

Minimum 8 characters

At least one uppercase letter

At least one lowercase letter

At least one digit

At least one special character (@, #, \$, %, &, !)

Mobile Number

Must contain exactly 10 digits.

First digit should be between 6–9.

Expected Output

Display:

Valid Email / Invalid Email

Strong Password / Weak Password

Valid Mobile Number / Invalid Mobile Number

Section 2: Design a Finite State Automata (FSA) Simulator

2. Problem Statement

Develop a Python-based Deterministic Finite Automata (DFA) simulator.

The program should:

- Accept the DFA description
 - States
 - Input alphabet
 - Transition table
 - Initial state
 - Final state(s)
- Accept multiple input strings.
- Simulate state transitions.
- Display whether the string is Accepted or Rejected.

Example

Language:

Strings ending with "ab"

Input

abaab

Output

Transition Path:

$q_0 \rightarrow q_1 \rightarrow q_0 \rightarrow q_1 \rightarrow q_2$

Accepted

Section 3: Smart Pattern Matching Engine using Regular Expressions**3. Problem Statement**

Develop a Python-based text search engine that performs pattern matching using Regular Expressions.

The system should support:

Word Search

Prefix Search

Suffix Search

Date Search

Phone Number Search

Hashtag Search

Mention (@username) Search

Example Input

Meeting on 12/09/2026

Call 9876543210

#NLP

@OpenAI

natural language processing

User Menu

1.Search Date

2.Search Phone Number

3.Search Hashtag

4.Search Mention

5.Search Prefix

6.Search Suffix

Expected Output

Display all matching patterns based on user selection.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Q. No. / Criteria	Problem Understanding	Algorithm	Code Implementation	Competitive Performance	Test Case Handling	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Section 1: Intelligent Email and Password Validator using Regular Expressions

1. Problem Statement:

A software company is developing a user registration system. Before creating an account, the system must validate user credentials based on predefined security policies.

Develop a Python application that validates:

Email Address

Password

Mobile Number

using Regular Expressions.

Validation Rules

Email

Must start with an alphabet.

May contain letters, digits, '.', '_' before '@'.

Domain should contain only alphabets.

Valid extensions: .com, .org, .edu, .net, .in

Password

Minimum 8 characters

At least one uppercase letter

At least one lowercase letter

At least one digit

At least one special character (@, #, \$, %, &, !)

Mobile Number

Must contain exactly 10 digits.

First digit should be between 6–9.

Expected Output

Display:

Valid Email / Invalid Email

Strong Password / Weak Password

Valid Mobile Number / Invalid Mobile Number

Code:-

```
import re

# User Input
email = input("Enter Email: ")
password = input("Enter Password: ")
mobile = input("Enter Mobile Number: ")

# Email Validation
email_pattern = r"^[A-Za-z][A-Za-z0-9-_.]*@[A-Za-z]+\.(com|org|edu|net|in)$"

if re.fullmatch(email_pattern, email):
    print("Valid Email")
else:
    print("Invalid Email")

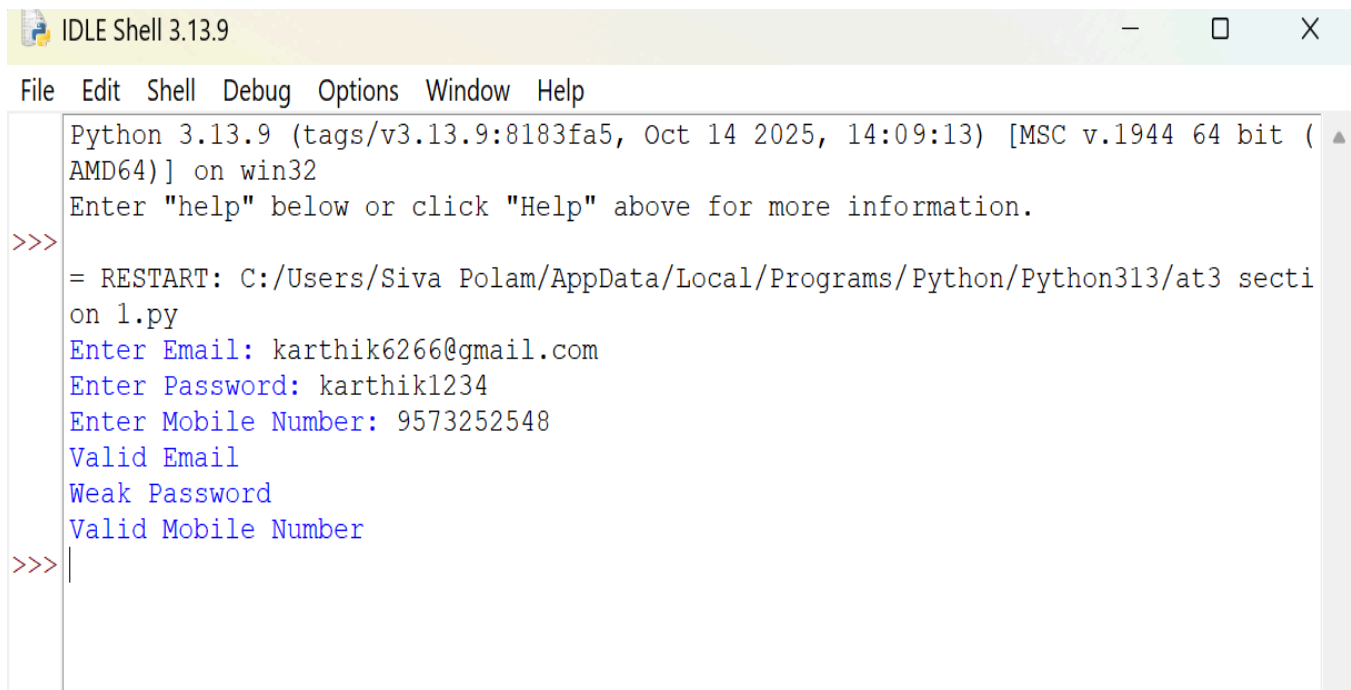
# Password Validation
password_pattern = r"^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#%&!])[A-Za-z\d@#%&!]{8,}$"
```

```
if re.fullmatch(password_pattern, password):
    print("Strong Password")
else:
    print("Weak Password")
```

```
# Mobile Number Validation
mobile_pattern = r"^[6-9]\d{9}$"
```

```
if re.fullmatch(mobile_pattern, mobile):
    print("Valid Mobile Number")
else:
    print("Invalid Mobile Number")
```

Output:



```
IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:/Users/Siva Polam/AppData/Local/Programs/Python/Python313/at3 section 1.py
Enter Email: karthik6266@gmail.com
Enter Password: karthik1234
Enter Mobile Number: 9573252548
Valid Email
Weak Password
Valid Mobile Number
>>> |
```

Section 2: Design a Finite State Automata (FSA) Simulator

4. Problem Statement

Develop a Python-based Deterministic Finite Automata (DFA) simulator.

The program should:

- Accept the DFA description
 - States
 - Input alphabet
 - Transition table
 - Initial state
 - Final state(s)
- Accept multiple input strings.
- Simulate state transitions.
- Display whether the string is Accepted or Rejected.

Example

Language:
Strings ending with "ab"

Input

abaab

Output

Transition Path:
 $q_0 \rightarrow q_1 \rightarrow q_0 \rightarrow q_1 \rightarrow q_2$
Accepted

Code:

```
# DFA for strings ending with "ab"

# Transition Table
dfa = {
    'q0': {'a': 'q1', 'b': 'q0'},
    'q1': {'a': 'q1', 'b': 'q2'},
    'q2': {'a': 'q1', 'b': 'q0'}
}

start_state = 'q0'
final_states = ['q2']

# Number of input strings
n = int(input("Enter number of strings: "))

for i in range(n):
    string = input("Enter string: ")
```

```

state = start_state
path = [state]

# Simulate DFA
for ch in string:
    if ch in dfa[state]:
        state = dfa[state][ch]
        path.append(state)
    else:
        print("Invalid Input")
        break
else:
    print("Transition Path:")
    print(" → ".join(path))

    if state in final_states:
        print("Accepted")
    else:
        print("Rejected")

print()

```

Output:

```

IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:/Users/Siva Polam/AppData/Local/Programs/Python/Python313/at3 section 2.py
Enter number of strings: 3
Enter string: abaab
Transition Path:
q0 → q1 → q2 → q1 → q1 → q2
Accepted

Enter string: abab
Transition Path:
q0 → q1 → q2 → q1 → q2
Accepted

Enter string: abb
Transition Path:
q0 → q1 → q2 → q0
Rejected

```


Section 3: Smart Pattern Matching Engine using Regular Expressions

1. Problem Statement

Develop a Python-based text search engine that performs pattern matching using Regular Expressions.

The system should support:

Word Search
Prefix Search
Suffix Search
Date Search
Phone Number Search
Hashtag Search
Mention (@username) Search

Example Input

Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing

User Menu

1.Search Date
2.Search Phone Number
3.Search Hashtag
4.Search Mention
5.Search Prefix
6.Search Suffix

Expected Output

Display all matching patterns based on user selection.

Code:

```
import re

# Sample Text
text = """
Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing
"""

while True:
    print("\n----- Smart Pattern Matching Engine -----")
    print("1. Search Date")
    print("2. Search Phone Number")
    print("3. Search Hashtag")
```

```
print("4. Search Mention")
print("5. Search Prefix")
print("6. Search Suffix")
print("7. Exit")

choice = int(input("Enter your choice: "))

if choice == 1:
    result = re.findall(r"\b\d{2}^\d{2}^\d{4}\b", text)
    print("Date:", result)

elif choice == 2:
    result = re.findall(r"\b[6-9]\d{9}\b", text)
    print("Phone Number:", result)

elif choice == 3:
    result = re.findall(r"#\w+", text)
    print("Hashtag:", result)

elif choice == 4:
    result = re.findall(r"@ \w+", text)
    print("Mention:", result)

elif choice == 5:
    prefix = input("Enter Prefix: ")
    result = re.findall(r"\b" + prefix + r"\w*", text, re.IGNORECASE)
    print("Matching Words:", result)

elif choice == 6:
    suffix = input("Enter Suffix: ")
    result = re.findall(r"\b\w*" + suffix + r"\b", text, re.IGNORECASE)
    print("Matching Words:", result)

elif choice == 7:
    print("Program Ended")
    break

else:
    print("Invalid Choice")
```

Output:

```
*IDLE Shell 3.13.9*
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Siva Polam/AppData/Local/Programs/Python/Python313/section 3.py

----- Smart Pattern Matching Engine -----
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Exit
Enter your choice: 1
Date: ['12/09/2026']

----- Smart Pattern Matching Engine -----
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Exit
Enter your choice: 2
Phone Number: ['9876543210']

----- Smart Pattern Matching Engine -----
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Exit
Enter your choice: 3
Hashtag: ['#NLP']
```